WORKED SOLUTION

STEP III 1996, Question 4

Pure • Maximal Product Partitions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q4

The key idea

Instead of searching through every way of splitting N, ask what an *optimal* splitting cannot contain. Any part that is too big can be broken into two pieces without lowering the product, and that is exactly what the opening inequality measures. Parts equal to 1 are wasteful, and too many 2s can be traded for 3s. After these moves only a handful of shapes survive, and the sum N picks out one of them.

Solution

The inequality

We have

$$k \leq 2(k-2) \iff k \leq 2k-4 \iff k \geq 4.$$

So the integers satisfying the inequality are $k = 4, 5, 6, \ldots$. Moreover there is equality only when $k = 4$, and for $k \geq 5$ the inequality is strict: $k < 2(k-2)$.

What an optimal splitting looks like

Fix $N \geq 2$. There are only finitely many ways of writing N as a sum of positive integers, so a largest product $P(N)$ exists. The order of the parts doesn't affect the sum or the product, so we treat a splitting as a collection of parts. Call a splitting *optimal* if its product is $P(N)$. We make three observations, each of which changes a splitting while keeping its sum equal to N.

(a) **No part is 5 or more.** If a part $m \geq 5$ is replaced by the two parts 2 and $m-2$ (both positive), the product is multiplied by $2(m-2)/m$, which is greater than 1 by the strict form of the inequality. So a splitting with a part $m \geq 5$ is not optimal.

A part equal to 4 can be replaced by $2+2$ with no change to the product, since $2 \times 2 = 4$ (the equality case $k = 4$). So among the optimal splittings there is one with no part bigger than 3. From now on we work with such an optimal splitting.

(b) **No part is 1.** Since $N \geq 2$, a part 1 is not the only part, so there is some other part a. Replacing the two parts 1 and a by the single part $a+1$ keeps the sum and changes the factor $1 \times a = a$ into $a+1 > a$. That increases the product, which is impossible for an optimal splitting.

(c) **At most two parts are 2.** If there were three 2s, replacing $2+2+2$ by $3+3$ would change the factor 8 into 9, again increasing the product.

So some optimal splitting consists of a threes and b twos with $b \in \{0,1,2\}$, and

$$N = 3a + 2b, \qquad P(N) = 3^a\, 2^b.$$

Ask yourself Could two different choices of (a,b) both fit the same N, so that we'd still have to compare them?

No. Reducing modulo 3 gives $2b \equiv N \pmod 3$, and the values $b = 0,1,2$ give $2b \equiv 0,2,1 \pmod 3$, three different residues. So N fixes b, and then $a = (N-2b)/3$ is fixed as well. Hence

$$P(N) = \begin{cases} 3^{N/3} & \text{if } N \equiv 0 \pmod 3, \\ 2 \times 3^{(N-2)/3} & \text{if } N \equiv 2 \pmod 3, \\ 2^2 \times 3^{(N-4)/3} & \text{if } N \equiv 1 \pmod 3,\ N \geq 4. \end{cases}$$

The values for $N = 5, \ldots, 9$

Applying this to each N in turn:

$$\begin{aligned} 5 &= 3+2, & P(5) &= 2 \times 3, \\ 6 &= 3+3, & P(6) &= 3^2, \\ 7 &= 3+2+2, & P(7) &= 2^2 \times 3, \\ 8 &= 3+3+2, & P(8) &= 2 \times 3^2, \\ 9 &= 3+3+3, & P(9) &= 3^3, \end{aligned}$$

as required. (For $N = 7$ the splitting $4+3$ is also optimal, which is why observation (a) only lets us *replace* a 4 rather than rule it out.)

Finding $P(1000)$

Since $1000 = 3 \times 332 + 4$, we have $1000 \equiv 1 \pmod 3$, so $b = 2$ and $a = (1000-4)/3 = 332$. The optimal splitting is 332 threes and two 2s, and

$$P(1000) = 2^2 \times 3^{332}.$$

Equivalently, the optimal splittings are 332 threes with two 2s, or 332 threes with one 4. These are the only ones, because the moves in (a)–(c) show that any other splitting can be strictly improved.

Checks

- A brute-force search over every splitting of N for $N \leq 40$ agrees with the formula, including the five given values.
- An exact computer calculation using $P(n) = \max_{1 \leq p \leq n} p\,P(n-p)$ gives $P(1000) = 4 \times 3^{332}$, a number with 160 digits.
- As a sanity check, compare with a non-integer "split": with N/x equal parts the product is $x^{N/x}$, which is largest at $x = \mathrm{e} \approx 2.72$. That is why 3s (and the odd 2) win.

Common slips

- Writing $1000 = 3 \times 333 + 1$ and answering 3^{333}: that splitting contains a 1, and $3 + 1$ should be replaced by $2 + 2$, since $4 > 3$.
- Saying that a part 4 "cannot occur". It can occur in an optimal splitting; it just never does better than $2 + 2$.
- Listing the values for $N = 5, \ldots, 9$ without showing that nothing beats them. The structure argument is what proves they are the largest.

Beyond STEP

The argument here is a typical *extremal* or *exchange* argument: assume a best object exists, then show that it must avoid certain local patterns. The same idea turns up in combinatorics, in proving that greedy algorithms are optimal, and in the theory of integer partitions. The continuous version (maximising $x^{N/x}$) is a pleasant calculus exercise. For more problems in this spirit see Siklos, *Advanced Problems in Mathematics*, and for congruences and elementary number theory, Davenport, *The Higher Arithmetic*.

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